

$V = \text{smooth var} / \mathbb{F} = \text{alg closure of } \mathbb{F}_q \Rightarrow \Gamma$

$$H_c^*(V) := \sum_{i \geq 0} (-1)^i [H_c^i(V, K)]_\Gamma \in K_0(K\Gamma)$$

Thm A.2.7

(a) If U is a Γ -stable open, $Z = V \setminus U$, then

$$H_c^*(V) = H_c^*(U) + H_c^*(Z)$$

$$Tr_V^* = Tr_U^* + Tr_Z^*$$

(b) If Δ is a finite subgroup of Γ normalized by Γ , then

$$H_c^*(V/\Delta) = H_c^*(V)^\Delta \text{ as elements of } K_0(K\Gamma/\Delta)$$

(c) If V' is another such variety with Γ -action

$$H_c^*(V \times V') = H_c^*(V) \otimes H_c^*(V')$$

(d) If s and u are two elements of Γ s.t. $su = us$, s is of order prime to p and u is of p -power order, then

$$Tr_V^*(su) = Tr_V^*(u)$$

(e) If Γ is a finite group and S is a torus over $\mathbb{F}$ (i.e. $S \cong (\mathbb{F}^\times)^n$) acting on V by regular apts and commuting with Γ , then

$$H_c^*(V) = H_c^*(V^S)$$

Recall/Learn A q -Weil number of weight $j \in \mathbb{N}$ is an algebraic integer α s.t. for any field embedding $\iota: \mathbb{Q}(\alpha) \hookrightarrow \mathbb{C}$, $|\iota(\alpha)| = q^{j/2}$

Thm Assume V is defined / $\mathbb{F}_q$, so F acts on V , hence on $H_c^i(V, K)$. Then

$$(a) Tr_V^*(F) = |V^F| = |V(\mathbb{F}_q)| \text{ (Grothendieck-Lefschetz fixed pt formula)}$$

(b) If V is irred of $\dim = d$, then F acts on $H_c^{2d}(V, K)$ by mult by q^d

(c) The eigenvalues of F on $H_c^i(V, K)$ are q -Weil numbers of weight $\leq i$

(d) If V is projective, the eigenvalues of F on $H_c^i(V, K)$ are q -Weil numbers of weight i .

Eg 1 $\mathbb{P}^1(\mathbb{F})$. Let $U = \{[x:y] \in \mathbb{P}^1(\mathbb{F}) : y \neq 0\} \cong \mathbb{A}^1(\mathbb{F})$

$$Z = \mathbb{P}^1(\mathbb{F}) \setminus U = \{\infty\}. \text{ Thm A.2.1} \Rightarrow$$

$$0 \rightarrow H_c^0(\mathbb{A}^1(\mathbb{F}), \Lambda) \rightarrow H_c^0(\mathbb{P}^1(\mathbb{F}), \Lambda) \rightarrow H_c^0(\{\infty\}, \Lambda)$$

$$\rightarrow H_c^1(\mathbb{A}^1(\mathbb{F}), \Lambda) \rightarrow H_c^1(\mathbb{P}^1(\mathbb{F}), \Lambda) \rightarrow H_c^1(\{\infty\}, \Lambda)$$

$$\rightarrow H_c^2(\mathbb{A}^1(\mathbb{F}), \Lambda) \rightarrow H_c^2(\mathbb{P}^1(\mathbb{F}), \Lambda) \rightarrow H_c^2(\{\infty\}, \Lambda)$$

$$\Rightarrow H_c^i(\mathbb{P}^1(\mathbb{F}), \Lambda) = \begin{cases} \Lambda & \text{if } i=0,2 \\ 0 & \text{else} \end{cases}$$

$$Tr_{\mathbb{P}^1(\mathbb{F})}^*(F) = 1 - 0 + q = q+1 = |\mathbb{P}^1(\mathbb{F}_q)|$$

Also $GL_2(\mathbb{F}) \curvearrowright \mathbb{P}^1(\mathbb{F}) \Rightarrow GL_2(\mathbb{F})$ acts trivially on $H_c^i(\mathbb{P}^1/\mathbb{F}, K)$.

Eg 2 $G_m(\mathbb{F}) = \mathbb{A}^1(\mathbb{F}) \setminus \{0\} = \mathbb{F}^\times$

Let $U = \mathbb{A}^1(\mathbb{F}) \setminus \{0\}$, $Z = \mathbb{A}^1(\mathbb{F}) \setminus U = \{0\}$, then we have

$$0 \rightarrow H_c^0(U, \Lambda) \rightarrow H_c^0(\mathbb{A}^1(\mathbb{F}), \Lambda) \rightarrow H_c^0(\{0\}, \Lambda)$$

$$\rightarrow H_c^1(U, \Lambda) \rightarrow H_c^1(\mathbb{A}^1(\mathbb{F}), \Lambda) \rightarrow H_c^1(\{0\}, \Lambda)$$

$$\rightarrow H_c^2(U, \Lambda) \rightarrow H_c^2(\mathbb{A}^1(\mathbb{F}), \Lambda) \rightarrow H_c^2(\{0\}, \Lambda)$$

$$\text{So } H_c^i(G_m(\mathbb{F}), \Lambda) \cong \begin{cases} \Lambda & \text{if } i=1,2 \\ 0 & \text{else} \end{cases}$$

$$\text{And } Tr_{G_m(\mathbb{F})}^*(F) = 0 - 1 + q = q-1 = |G_m(\mathbb{F}_q)|$$

Also the action of $G_m(\mathbb{F}) = \mathbb{F}^\times$ on itself is the trivial action on $H_c^i(G_m(\mathbb{F}), K)$.

§ Deligne-Lusztig induction

Back to $G = SL_2(\mathbb{F}_q)$

$$\cdot Y = \{xy^q - yx^q = 1\} \supset G \times \mu_{q+1}$$

$\cdot K/\mathbb{Q}_\ell$ fin ext containing all $|G|$ -roots of 1.

$\leadsto H_c^i(Y) := H_c^i(Y, K)$ a $(KG, K\mu_{q+1})$ -bimodule

Def If Θ is a character of μ_{q+1} and V_Θ is a rep with trace char Θ , we define

$$R'(\Theta) := - \sum_{i \geq 0} (-1)^i [H_c^i(Y) \otimes_{K\mu_{q+1}} V_\Theta]_G$$

This extends to a hom

$$R': \mathcal{K}_o(K_{\mu_{q+1}}) \rightarrow \mathcal{K}_o(KG)$$

This is called Deligne-Lusztig induction.